

Course Description

Course Name	Applied Electronics																
Course Code																	
Competencies	The students pay particular attention to the following competencies: - Identify, test, and characterize electronic components using standard instruments - Build and troubleshoot analog and digital circuits on breadboard and PCB - Operate laboratory instruments including multimeters, oscilloscopes, and function generators - Analyze and verify the behavior of amplifier, filter, and logic circuits - Integrate sensors, microcontrollers, and actuators into functional IoT-based embedded systems																
Related World Skill Competition	Electronics																
Course Outcomes	After completion of the course, the student will be able to: CO1: Identify and characterize electronic components including resistors, capacitors, inductors, diodes, transistors, and ICs using color codes, datasheets, and measurement instruments. CO2: Operate electronic laboratory instruments such as multimeters, oscilloscopes, and function generators to measure, generate, and analyze electrical signals including AC/DC quantities, waveform parameters, and frequency. CO3: Design, assemble, and debug electronic circuits on breadboard and PCB, including LED circuits, voltage dividers, and rectifiers, using standard circuit construction and PCB layout techniques. CO4: Analyze, build, and verify analog circuits such as amplifiers and filters and digital circuits including logic gates, flip-flops, and counters, confirming behavior through measurement and truth table verification. CO5: Interface sensors, microcontrollers, and actuators to develop basic IoT-based embedded systems and transmit data to an IoT platform using wireless communication protocols.																
Course Structure	<table border="1"> <thead> <tr> <th>Total Contact Hrs. (LTP)</th><th>Days</th><th>Total L&T Hrs.</th><th>Total P Hrs.</th><th>Total SL Hrs.</th><th>Credits</th></tr> </thead> <tbody> <tr> <td>90</td><td>15-18</td><td>30</td><td>60</td><td>90</td><td>4</td></tr> </tbody> </table>					Total Contact Hrs. (LTP)	Days	Total L&T Hrs.	Total P Hrs.	Total SL Hrs.	Credits	90	15-18	30	60	90	4
Total Contact Hrs. (LTP)	Days	Total L&T Hrs.	Total P Hrs.	Total SL Hrs.	Credits												
90	15-18	30	60	90	4												
Assessment Scheme																	

Component	Sub-Component	Total Assessments	Marks (Max. 100)
Continuous Assessment (CA)	Assignments	2	20
	Practical Exams	2	40
	Report Writing	1	10
	Sub Total	5	70
Terminal Assessment (TA)	Written Exams	1	30
	Sub Total	1	30
Reference Books			
1. Boylestad, R. L. (2009). <i>Electronic devices and circuit theory</i> . Pearson Education India. 2. https://docs.arduino.cc/ 3. Floyd, T. L. (2005). <i>Electronic devices</i> . Pearson Education India.			

Detailed Syllabus

Unit	Name	Content	No. of Hrs.
I	Electronic Components & Basic Measurements	K1: Active vs. passive components — resistors, capacitors, inductors, diodes, transistors, and ICs. K2: Component identification using color codes, markings, and manufacturer datasheets. K3: Introduction to measurement techniques: voltage, current, resistance, and continuity. K4: Diode and transistor characteristics and testing principles. K5: Capacitor and inductor identification, value reading, and testing methods.	4
		A1: Identifying resistors, capacitors, inductors, diodes, and transistors from a component kit. A2: Measuring resistance, DC voltage, and current using a digital multimeter. A3: Testing diodes for forward/reverse bias and transistors for gain and functionality. A4: Verifying capacitor and inductor values using an LCR meter. A5: Reading and interpreting component datasheets for specifications and ratings.	14
II	Electronic Instruments & Signal Analysis	K1: Operating principles and specifications of the digital multimeter.	4

		K2: Oscilloscope architecture, controls, probing techniques, and measurement modes. K3: Function generator operation, output impedance, and waveform selection. K4: Waveform parameters: frequency, period, amplitude, peak-to-peak, and phase. K5: Comparison of sine, square, and triangle waveforms and their applications.	
		A1: Generating sine, square, and triangle waveforms using a function generator. A2: Observing and measuring waveforms on an oscilloscope (time/div, volt/div). A3: Measuring AC and DC voltages and frequency with the oscilloscope. A4: Analyzing frequency response and amplitude variations across different load conditions. A5: Comparing measured waveform parameters with function generator settings.	14
III	Circuit Implementation (Breadboard & PCB)	K1: Breadboard structure, connection layout, and current path principles. K2: Circuit assembly techniques: component placement, wire routing, and bus conventions. K3: Introduction to PCB design: layers, traces, pads, vias, and design rules. K4: PCB fabrication process overview: etching, drilling, and soldering.	4
		A1: Building LED indicator, voltage divider, and half-wave/full-wave rectifier circuits on breadboard. A2: Verifying circuit operation by measuring output voltages and comparing to calculated values. A3: Designing a simple single-layer PCB layout using EDA software (e.g., KiCad or EasyEDA). A4: Assembling and soldering a basic PCB and testing functionality.	14
IV	Analog & Digital Circuit Analysis	K1: Amplifier fundamentals: voltage gain, biasing, and operating point. K2: Filter types — low-pass, high-pass, band-pass — and their frequency characteristics. K3: Logic gates: AND, OR, NOT, NAND, NOR, XOR operations and truth tables. K4: Flip-flop types (SR, D, JK) and their role in sequential logic. K5: Counter circuits: asynchronous and synchronous counting operation.	4
		A1: Building and testing a common-emitter amplifier; measuring voltage gain with oscilloscope. A2: Constructing low-pass and high-pass RC filter circuits and verifying frequency response.	14

		A3: Building basic logic gate circuits (AND, OR, NOT, NAND) and verifying truth tables. A4: Implementing a D flip-flop circuit and demonstrating clocked operation. A5: Building a 4-bit ripple counter and verifying binary counting sequence.	
V	Sensors, Controllers, IoT Systems	K1: Sensor types and operating principles: temperature (LM35/DHT11), light (LDR), and motion (PIR). K2: Microcontroller architecture (Arduino), GPIO, ADC, and digital I/O concepts. K3: Actuator : LEDs,Speakers K4: IoT system fundamentals: data acquisition, communication protocols, and cloud basics. K5: Wireless communication modules: Wi-Fi (ESP8266) and Bluetooth (HC-05) integration.	4
		A1: Interfacing temperature and light sensors with Arduino and displaying readings via serial monitor. A2: Controlling a DC motor and relay using digital output pins. A3: Building a smart light system that responds automatically to ambient light level. A4: Building a temperature monitoring system that logs and displays data. A5: Transmitting sensor data to an IoT platform (e.g., Blynk or ThingSpeak) via Wi-Fi or Bluetooth.	14
Total			90

Examples of Practical Exercises

Exercise 1: Component Identification and Testing

Given Scenario:

A set of unlabeled electronic components is provided in a lab tray.

Students will perform the following:

- Identify resistors, capacitors, inductors, diodes, and transistors visually and using color codes.
- Measure resistance, voltage, and current using a digital multimeter.
- Test diodes in forward and reverse bias modes.
- Test transistor gain using the multimeter's hFE function.
- Record measured values and compare with datasheet specifications.

Exercise 2: Signal Measurement Using Oscilloscope and Function Generator

Given Scenario:

A technician needs to verify the output of a signal source before connecting it to a circuit.

Students will perform the following:

- Set up the function generator to produce a 1 kHz, 5 V peak-to-peak sine wave.
- Connect the oscilloscope probe correctly and observe the waveform.
- Measure frequency, period, amplitude, and peak-to-peak voltage.
- Switch waveforms to square and triangle and repeat measurements.
- Document observed values and explain any deviations from set parameters.

Exercise 3: Breadboard Circuit Construction and PCB Design

Given Scenario:

A simple power supply circuit must be prototyped and then transferred to a PCB.

Students will perform the following:

- Build a full-wave bridge rectifier circuit on a breadboard and measure the DC output.
- Add a filter capacitor and observe the effect on ripple voltage.
- Design a PCB layout for the rectifier using EDA software.
- Export the Gerber files and review design rule check (DRC) results.
- Assemble and solder the basic PCB, then verify output with a multimeter.

Exercise 4: Analog and Digital Circuit Verification

Given Scenario:

An electronics technician must verify the gain of an amplifier and the operation of a combinational logic circuit.

Students will perform the following:

- Build a common-emitter amplifier and apply a sinusoidal input signal.
- Measure input and output voltages using the oscilloscope and calculate voltage gain.
- Build a 3-input logic circuit and verify all possible input combinations against the truth table.
- Implement a D flip-flop and verify clocked behavior.
- Document results and identify any discrepancies between expected and measured outputs.

Exercise 5: IoT-Based Embedded System Integration

Given Scenario:

An organization needs a smart monitoring system that can log environment data remotely.

Students will perform the following:

- Interface a DHT11 temperature and humidity sensor with an Arduino microcontroller.
- Display sensor readings on the serial monitor and an LCD screen.
- Program an automatic relay to trigger based on temperature threshold.
- Configure Wi-Fi module (ESP8266) and connect to an IoT platform (ThingSpeak or Blynk).
- Transmit live sensor data and verify it appears correctly on the IoT dashboard.

Examples of Problem-Solving Tasks

Task 1: Circuit Does Not Produce Expected Output

Given Scenario:

A student builds a rectifier circuit but measures 0 V at the output.

Student will:

- Identify possible causes such as incorrect component placement, a faulty diode, or a broken breadboard connection.
- Use a multimeter to test each diode in circuit for proper junction behavior.
- Verify the power supply voltage at the input.
- Trace signal flow from input to output using a systematic probing approach.
- Apply the corrective action and verify the expected DC output is produced.

Task 2: Oscilloscope Displays Distorted or Unstable Waveform

Given Scenario:

An oscilloscope display shows a waveform that is unstable and continuously shifting.

Student will:

- Identify possible causes including incorrect triggering settings, poor probe grounding, or unstable input signal.
- Adjust trigger level, trigger source, and coupling mode on the oscilloscope.
- Check probe ground connection and probe compensation.
- Verify function generator output is stable and at the correct frequency.
- Stabilize the waveform display and explain the corrective adjustments made.

Task 3: Logic Circuit Produces Incorrect Output

Given Scenario:

A logic gate circuit built on a breadboard does not produce the expected output for certain input combinations.

Student will:

- Compare the physical circuit wiring against the schematic diagram.
- Use a logic probe or multimeter to check logic levels at each gate input and output.
- Identify any floating inputs, missing power/ground connections, or incorrect IC pin assignments.
- Correct the circuit and verify the complete truth table.
- Justify the observed fault and explain the logic error in the original construction.

Task 4: Microcontroller Sensor Reading Is Inaccurate or Absent

Given Scenario:

An Arduino connected to a temperature sensor outputs constant zero or erratic values on the serial monitor.

Student will:

- Check sensor wiring: VCC, GND, and signal pin connections.
- Verify the correct library is included and pin assignment matches the code.
- Measure sensor output voltage directly with a multimeter to confirm it is powered.
- Upload a simplified test sketch to isolate the fault between hardware and software.
- Correct the fault and demonstrate accurate live readings on the serial monitor.

Task 5: IoT Device Fails to Connect to Platform

Given Scenario:

A Wi-Fi-enabled sensor node is not transmitting data to the IoT cloud platform.

Student will:

- Check the Wi-Fi SSID and password in the code for correctness.

- Verify the Wi-Fi module is powered and physically connected correctly.
- Use the serial monitor to identify connection error messages from the module.
- Confirm the IoT platform API key and endpoint URL are correctly configured.
- Successfully establish the connection and verify live data appears on the IoT dashboard.

Examples of Problem-Solving Questions

Q1. A company is setting up a quality control station that must continuously monitor ambient temperature and trigger an alarm when values exceed a safe threshold. The team is deciding between using a standalone analog comparator circuit and a microcontroller-based system.

Compare both approaches in terms of accuracy, programmability, cost, and maintainability. Evaluate the risk of using only an analog comparator in a dynamic industrial environment. Recommend the more suitable solution and justify your answer based on reliability and long-term operational requirements.

Q2. A technician must choose between constructing a prototype on a breadboard and fabricating a printed circuit board (PCB) for a voltage regulator circuit that will be used in a product assembly line.

Compare breadboard prototyping and PCB fabrication in terms of reliability, speed of assembly, noise susceptibility, and suitability for production. Identify the risks of deploying a breadboard-based circuit in a production environment. Which approach would you recommend for the assembly line and why?

Q3. An engineering team is developing a data logger that records temperature, humidity, and light levels every 30 seconds and stores the readings in the cloud. The system must run continuously for months without manual intervention.

Analyze the sensor selection, microcontroller power management, and data transmission strategy needed for such a system. Compare continuous transmission versus periodic batch transmission in terms of power consumption and reliability. Propose a suitable architecture and justify your design decisions.

Q4. A digital control system based on combinational logic gates is producing incorrect output states. The design engineer must decide whether to debug the existing circuit or replace it with a microcontroller-based solution.

Compare hardwired combinational logic with a software-based microcontroller implementation in terms of flexibility, debugging ease, cost, and scalability. Evaluate the long-term consequences of maintaining a complex fixed logic circuit. Recommend the more appropriate approach and justify your reasoning.

Q5. An electronics lab is evaluating whether students should use analog oscilloscopes or digital storage oscilloscopes (DSO) for signal analysis tasks involving both periodic waveforms and single-shot events.

Compare analog and digital oscilloscopes in terms of bandwidth, triggering capability, waveform storage, and ease of use for different measurement scenarios. Justify which type of oscilloscope provides greater educational value for an applied electronics course and explain the impact of instrument choice on measurement accuracy and student learning outcomes.